

Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C , such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$. Then, there exists rational maps $\pi, \pi' : X \dashrightarrow C, C'$ and sections $\sigma, \sigma' : C, C' \rightarrow X$ such that $\pi \circ \sigma$ and $\pi' \circ \sigma'$ extend to the identity morphism on C and C' , respectively. Then $\pi' \circ \sigma : C \rightarrow C'$ and $\pi \circ \sigma' : C' \rightarrow C$ are inverses to each other, so we have an isomorphism $C \cong C'$. $\square$

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C . Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. If $\mathcal{E}$ is decomposable, then $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$, where $\mathcal{L} = \mathcal{O}_C(\mathfrak{e})$ for some divisor $\mathfrak{e} \in \text{Div } C$. By (2.8.1), (2.11.1), and (2.11.2), the two natural projections $\mathcal{E} \rightarrow \mathcal{L}$ and $\mathcal{E} \rightarrow \mathcal{O}_C$ correspond to sections $C' \sim C_0$ and $C'' \sim C_0 - \mathfrak{e}f$, respectively, such that $C' \cdot C'' = 0$. Hence, $C' \cap C'' = \emptyset$. Conversely, suppose there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$. The corresponding surjective morphisms $\mathcal{E} \rightarrow \mathcal{L}'$ and $\mathcal{E} \rightarrow \mathcal{L}''$ induce an $\mathcal{O}_C$ -module homomorphism $\mathcal{E} \rightarrow \mathcal{L}' \oplus \mathcal{L}''$. For every $P \in C$, the morphism of stalks $\mathcal{E}_P \rightarrow \mathcal{L}'_P \oplus \mathcal{L}''_P$ is an isomorphism because C' and C'' are disjoint, so $\mathcal{L}'_P \oplus \mathcal{L}''_P$ is generated by $\mathcal{E}_P$. Hence, $\mathcal{E} \rightarrow \mathcal{L}' \oplus \mathcal{L}''$ is an isomorphism. $\square$

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C , then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \dots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

- (b) Show that this is false for varieties of dimension ≥ 2 . In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

- (a) We proceed by induction on r . The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{L}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i/\mathcal{E}_{i-1} \cong (\mathcal{E}'_i/\mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \dots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{F}$ is also globally generated locally free of rank $r - 1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{F}_i)$ of $\mathcal{F}$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{F}_i)$, where we relabel $j = i + 1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \text{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j/\mathcal{E}_{j-1} \cong \mathcal{F}_{j-1}/\mathcal{F}_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

- (b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $H^1(\mathcal{L}), H^1(\mathcal{M}) = 0$, so by the long exact of cohomology, $H^1(\Omega) = 0$. But by (III, Ex. 7.2), $H^1(\Omega) = k$, a contradiction.

□

4. Let C be a curve of genus g , and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

- (a) Show that $r = 0$ and $r \geq g + 1$ are always possible.
- (b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \rightarrow \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0 \cdot D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

- (a) There is a one-to-one correspondence between sections D of X and finite separable morphisms $f : C \rightarrow \mathbb{P}^1$. In particular, a section D is the graph of a morphism $f : C \rightarrow \mathbb{P}^1$, and its self-intersection is $D^2 = 2 \cdot \deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geq g + 1$ is always possible.
- (b) If $g > 0$, then there is no morphism $C \rightarrow \mathbb{P}^1$ of degree 1. If C is a hyperelliptic curve, then it has a g_2^1 , so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g_3^1 (IV, 5.5.2), so $r = 3$ is always possible.

□

5. *Values of e .* Let C be a curve of genus $g \geq 1$.

- (a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.
- (b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(\mathcal{O}(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

- (c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e .
- (d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X .

Proof.

- (a) Let $\mathfrak{e}$ be a divisor of degree $-e$ on C , and let $\mathfrak{k}$ is a canonical divisor of C . We need to first construct an indecomposable extension $\mathcal{E}$ of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$. By (III, Ex. 6.1), an extension of vector bundles

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(\mathfrak{e}) \longrightarrow 0$$

corresponds to an element in $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. In this case, $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O}) \cong H^1(\mathcal{O}(-\mathfrak{e}))$ (III, 6.3) and (III, 6.7). This is dual to $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$, so it is enough to find $\mathfrak{e}$ such that $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$. The image of the canonical morphism $C \rightarrow |\mathfrak{k}| \cong \mathbb{P}^{g-1}$ is either a rational curve of degree $g - 1$, or an embedding of degree $2g - 2$, depending on whether or not C is hyperelliptic. If $-\mathfrak{e}$ is effective, then $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$ is the space of all hyperplane sections in $\mathbb{P}^{g-1}$ with support containing $\mathfrak{e}$. In any case, a hyperplane section of C in $\mathbb{P}^{g-1}$ consists of $2g - 2$ points counted with multiplicity. Hence, for each $0 \leq e \leq 2g - 2$, there is a divisor $\mathfrak{e}$ of degree $-e$ such that $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$, so there exists indecomposable extensions of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$.

Now, let $\mathcal{E}$ be an extension of $\mathcal{O}(\mathfrak{e})$ defined by a nonzero element in $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. It remains to show $\mathcal{E}$ is normalized and indecomposable. Clearly $H^0(\mathcal{E}) \neq 0$ by construction. If $\mathcal{M}$ is any invertible sheaf, then we have an exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(\mathfrak{e}) \otimes \mathcal{M} \longrightarrow 0.$$

If $\deg \mathcal{M} < 0$, then we have $H^0(\mathcal{M}) = 0$ and $H^0(\mathcal{O}(\mathfrak{e}) \otimes \mathcal{M}) = 0$, and therefore also $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$. Furthermore, if $\mathcal{E}$ is decomposable, then it would be $\mathcal{O} \oplus \mathcal{L}$ for some $\mathcal{L}$, by (2.12). But $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\mathfrak{e})$, so $\mathcal{L} \cong \mathcal{O}(\mathfrak{e})$, so in fact this extension would have to split, which it doesn't. Hence, for each $0 \leq e \leq 2g - 2$, there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.

- (b) Let $\xi \in H^1(\mathcal{O}(-D))$ be the (non-zero) element defining the extension $\mathcal{E}$. If E is any effective divisor of degree $d - 1$ on C , then we have a commutative diagram

$$\begin{array}{ccc} H^0(\mathcal{O}) & \xrightarrow{\delta} & H^1(\mathcal{O}(-D)) \\ \alpha \downarrow & & \beta \downarrow \\ H^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & H^1(\mathcal{O}(E - D)) \end{array}$$

where $\delta(1) = \xi$, $\alpha(1) = t$, a non-zero section defining E , and β is induced from the map $\mathcal{O} \rightarrow \mathcal{O}(E)$ corresponding to t . On the other hand, β is dual to the map

$$\beta^\vee : H^0(\mathcal{O}(D + K - E)) \rightarrow H^0(\mathcal{O}(D + K))$$

also induced by t . The subspace of $H^0(\mathcal{O}(D + K))$ corresponding to L_E is precisely the image of $\beta^\vee$, so $\beta(\xi) = 0$ if and only if $L_E \subseteq H$.

If $\mathcal{E}$ is normalized, then γ is injective for any effective divisor E of degree $d - 1$. By the commutative diagram, we must have $\beta(\xi) \neq 0$. Hence, $L_E \not\subseteq H$.

Conversely, if $\mathcal{E}$ is not normalized, then $H^0(\mathcal{E} \otimes \mathcal{M}) \neq 0$ for some invertible sheaf $\mathcal{M}$ with $\deg \mathcal{M} < 0$, and let $m = -\deg \mathcal{M}$. We have the exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(D) \otimes \mathcal{M} \longrightarrow 0.$$

If $m \geq d$, then we have $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$ since $\deg \mathcal{O}(D) \otimes \mathcal{M} < 0$, and if $m = d$, then either $H^0(\mathcal{O}(D) \otimes \mathcal{M}) = 0$, or $\mathcal{M} = \mathcal{O}(-D)$, whence the coboundary map δ is injective, hence $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$. So we can assume $0 < m < d$. The long exact sequence of cohomology gives

$$0 \longrightarrow H^0(\mathcal{E} \otimes \mathcal{M}) \longrightarrow H^0(\mathcal{O}(D) \otimes \mathcal{M}) \longrightarrow H^1(\mathcal{M}),$$

where $H^0(\mathcal{M}) = 0$. The middle term is non-zero, so if $\mathcal{M} \cong \mathcal{O}(-M)$ with $m = \deg M$, then $D - M \sim E$ for some effective divisor E of degree $d - m$. For any effective divisor $\tilde{E}$ of degree $d - 1$ with $\tilde{E} - E \geq 0$, we have a commutative diagram,

$$\tilde{\alpha} \left(\begin{array}{ccc} H^0(\mathcal{O}) & \xrightarrow{\delta} & H^1(\mathcal{O}(-D)) \\ \downarrow \alpha & & \downarrow \beta \\ H^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & H^1(\mathcal{O}(E - D)) \\ \downarrow \phi & & \downarrow \psi \\ H^0(\mathcal{O}(\tilde{E})) & \xrightarrow{\eta} & H^1(\mathcal{O}(\tilde{E} - D)) \end{array} \right) \tilde{\beta},$$

where $\delta(1) = \xi$, α, β is induced by a section $t \in H^0(\mathcal{O}(E))$ defining E , and ϕ, ψ is induced by a section s defining $\tilde{E} - E$.

Now, if $\beta(\xi) = 0$, then we are done. Otherwise, dualizing the column on the right of the commutative diagram gives

$$\begin{array}{ccc} H^0(\mathcal{O}(D + K - \tilde{E})) & & \\ \downarrow \psi^\vee & \searrow \tilde{\beta}^\vee & \\ H^0(\mathcal{O}(D + K - E)) & \xrightarrow{\beta^\vee} & H^0(\mathcal{O}(D + K)) \end{array}.$$

Thus, for every $\tilde{E}$, we have $L_{\tilde{E}} \subseteq L_E$. By the Riemann-Roch theorem, we have

$$\dim H^0(\mathcal{O}(D + K - \tilde{E})) = g, \quad \dim H^0(\mathcal{O}(D + K - E)) = g + m - 1, \quad \dim H^0(\mathcal{O}(D + K)) = g + d - 1.$$

The space of possible $\tilde{E}$ has dimension $m - 1$, so by varying $\tilde{E}$, we get every divisor in the linear system $|D + K - E|$. Also, $\beta(\xi) \neq 0$ if and only if $L_E \not\subseteq H$. Consequently, for at least one $\tilde{E}$, we have $L_{\tilde{E}} \not\subseteq H$, which is what we wanted to show.

- (c) Let $\mathbb{P}^N = |K + D|$ with $N = g + d - 2$. For any effective divisor E of degree $d - 1$ on C , let us consider the set

$$B_E = \{\text{hyperplanes } H \subset \mathbb{P}^N \mid L_E \subseteq H\} \subset |H|,$$

where $|H|$ is the complete linear system of hyperplanes in $\mathbb{P}^N$, i.e., $|H| = |K + D|$. A hyperplane in $\mathbb{P}^N$ is determined by a nonzero global section $f \in V = H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. Viewing L_E as a linear subspace of $\mathbb{P}^N$, let $W_E = H^0(L_E, \mathcal{O}_{L_E}(1))$. The space of effective divisors of degree $d - 1$ is parameterized by the closed points of the symmetric product $C^{(d-1)}$, which has dimension $d - 1$. Consider the restriction map

$$\varphi_x : V \rightarrow W_E.$$

It is clearly surjective, and the hyperplanes $H \in B_E$ correspond exactly to those $f \in \ker \varphi_E$. We have $\dim V = g + d - 1$ and $\dim W_E = g$, so $\dim B_E = \dim(\ker \varphi_E) - 1 = d - 2$.

Now, consider the subset $B \subset C^{(d-1)} \times |H|$ consisting of all pairs (E, H) such that $E \in C^{(d)}$ is a closed point and $H \in B_E$. Clearly B is the set of closed points of a $C^{(d-1)} \times |H|$, which we denote also by B , and which we give a reduced induced scheme structure. The first projection $p_1 : B \rightarrow C^{(d-1)}$ is surjective, with fibre a projective space of dimension $d - 2$. So B has dimension $(d - 1) + (d - 2) = 2d - 3$, which is strictly smaller than $\dim |H| = g + d - 2$. If $H \in |H| - p_2(B)$, then $H \not\supseteq L_E$ for every $E \in C^{(d-1)}$, so the extension of $\mathcal{O}(D)$ by $\mathcal{O}$ given by the element in $H^0(\mathcal{O}(K + D)) \cong H^1(\mathcal{O}(-D))$ corresponding to H satisfies the requirement of (b). Hence, if $g \leq e < 0$, then there exists a ruled surface X over C with invariant e .

- (d) Let C be a curve of genus $g = 2$, let $\mathcal{E}$ be a normalized locally free sheaf of rank 2 on C with invariant e , let $d = -e$, and let $X = \mathbb{P}(\mathcal{E})$ be the ruled surfaces associated to $\mathcal{E}$. As in the proof of (2.15), we want to show $e = -3, -4$ does not occur.

If it did, we would have a normalized bundle $\mathcal{E}$ with an exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0,$$

where $D = P_1 + \dots + P_d$ with $d = 3, 4$, since by the Riemann-Roch theorem, every divisor of degree > 2 on C is linearly equivalent to an effective divisor. Now take any set of points $Q_1, \dots, Q_d \in C$ such that $Q_1 + \dots + Q_d \sim P_1 + \dots + P_d$, and let $\mathcal{M} = \mathcal{O}(-Q_1)$. Then, since $\mathcal{E}$ is normalized, $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$, the map $\gamma : H^0(\mathcal{O}(D - Q_1)) \rightarrow H^1(\mathcal{O}(-Q_1))$ must be injective. On the other hand, let $\xi \in H^1(\mathcal{O}(-D))$ be the element defining the extension $\mathcal{E}$. Then we have a commutative diagram, writing $\mathcal{O}(D - Q_1)$ as $\mathcal{O}(S)$,

$$\begin{array}{ccc} H^0(\mathcal{O}) & \xrightarrow{\delta} & H^1(\mathcal{O}(-D)) \\ \alpha \downarrow & & \beta \downarrow \\ H^0(\mathcal{O}(S)) & \xrightarrow{\gamma} & H^1(\mathcal{O}(-Q_1)) \end{array}$$

where $\delta(1) = \xi$, $\alpha(1) = t$, a section defining the divisor S , and β is induced from the map $\mathcal{O} \rightarrow \mathcal{O}(S)$ corresponding to t . Now β is dual to the map

$$\beta^\vee : H^0(\mathcal{O}(K + Q_1)) \rightarrow H^0(\mathcal{O}(K + D))$$

also induced by t . The image of any nonzero element of $H^0(\mathcal{O}(K + Q_1))$ by $\beta^\vee$ is a section of $H^0(\mathcal{O}(K + D))$ corresponding to the effective divisor $K + Q_1 + S \in |K + D|$. By the Riemann-Roch theorem, we have

$$\dim H^0(\mathcal{O}(K + Q_1)) = 2, \quad \dim H^0(\mathcal{O}(K + D)) = d + 1.$$

By varying Q_1 and S , we get every divisor in the linear system $|D|$.

□

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. We call any locally free sheaf $\mathcal{E}$ on a curve C *normalized* if $H^0(C, \mathcal{E}) \neq 0$ and $H^0(C, \mathcal{E} \otimes \mathcal{L}) = 0$ for all invertible sheaves $\mathcal{L}$ on C with $\deg \mathcal{L} < 0$. With this definition, the statement of (2.8) can be made sense for $\mathcal{E}$ of arbitrary rank:

Lemma 0.1. *For every locally free sheaf $\mathcal{E}$ of finite rank on a curve C , there exists an invertible sheaf $\mathcal{L}$ on C such that $\mathcal{E} \otimes \mathcal{L}$ is normalized.*

Going back to the original statement, we can assume $\mathcal{E}$ is normalized. We proceed by induction on the rank r of $\mathcal{E}$. The base case $r = 1$ is trivial, so assume the statement holds for any $< r$. Pick a subinvertible sheaf $0 \rightarrow \mathcal{O} \rightarrow \mathcal{E}$ of maximal degree, and consider the short exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

The quotient sheaf $\mathcal{F} = \mathcal{E}/\mathcal{O}$ is locally free of rank $r - 1$ because of the choice of $0 \rightarrow \mathcal{O} \rightarrow \mathcal{E}$ (2.8), so by the inductive hypothesis, $\mathcal{F}$ is decomposable, say $\mathcal{F} = \bigoplus \mathcal{O}(b_i)$ for some $b_i \in \mathbb{Z}$. Tensoring by $\mathcal{O}(-1)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathcal{O}(-1)) \longrightarrow H^0(\mathcal{E}(-1)) \longrightarrow H^0(\bigoplus \mathcal{O}(b_i - 1)) .$$

Since $\mathcal{E}$ is normalized, the middle vector space is zero, so by exactness we have $b_i \leq 0$. On the other hand, the space of extensions of $\mathcal{F}$ by $\mathcal{O}$ can be computed by Serre duality, which gives

$$\mathrm{Ext}^1(\mathcal{F}, \mathcal{O}) \cong H^0(\bigoplus \mathcal{O}(b_i - 1)) = 0,$$

so any extension of $\mathcal{F}$ by $\mathcal{O}$ is trivial. Hence, we have $\mathcal{E} \cong \mathcal{O} \oplus \mathcal{F}$. □

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C , and that no two are linearly equivalent.

Proof. Let C be an elliptic curve, let $P \in C$, and construct a locally free sheaf $\mathcal{E}$ of rank 2 as an extension

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(P) \longrightarrow 0$$

defined by a nonzero element $\xi \in \text{Ext}^1(\mathcal{O}(P), \mathcal{O}) \cong H^0(\mathcal{O}(P))$. By (2.11.6) and (2.15), $\mathcal{E}$ is normalized, and it is the unique ruled surface over C with invariant $e = -1$. If $\mathcal{E}'$ is similarly defined for a point $Q \in C$, then $\mathcal{E} \otimes \mathcal{O}(R - P) \cong \mathcal{E}'$, where $R \in C$ such that $2R \sim P + Q$. $\square$

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F}) / \text{rank } \mathcal{F} > (\deg \mathcal{E}) / \text{rank } \mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable*.

- (a) A decomposable $\mathcal{E}$ is never stable.
- (b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E} > 0$ (respectively, ≥ 0).
- (c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g - 2$, (2) an element $\mathcal{L} \in \text{Pic } C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(X, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.

Proof.

- (a)
- (b)
- (c)

$\square$

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.
10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g_3^1 .
11. Let X be a ruled surface over the curve C , defined by a normalized bundle $\mathcal{E}$, and let ϵ be the divisor on C for which $\mathcal{O}_C(\epsilon) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C .
- (a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \epsilon|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no points.
 - (b) If $\mathfrak{b}$ and $\mathfrak{b} + \epsilon$ are very ample on C , and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \epsilon - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

12. Let X be a ruled surface with invariant e over an elliptic curve C , and let $\mathfrak{b}$ be a divisor on C .

- (a) If $\deg \mathfrak{b} \geq e + 2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.
- (b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geq e + 3$.

Proof.

(a)

□

13. For every $e \geq -1$ and $n \geq e + 3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.

Proof.

□

14. Let X be a ruled surface over a curve of genus g , with invariant $e < 0$, and assume that $\text{char } k = p > 0$ and $g \geq 2$.

- (a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0.f$, then either $a = 1, b \geq 0$, or $2 \leq a \leq p-1, b \geq \frac{1}{2}ae$, or $a \geq p, b \geq \frac{1}{2}ae + 1 - g$.
- (b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

Proof.

- (a)
- (b)

□

15. *Funny behaviour in characteristic p .* Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.

- (a) Show that the action of the k -linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.
- (b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.
- (c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C . Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi : Y \rightarrow C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a Dedekind domain that is finitely generated over k such that $C \cong \text{Spec } A$. The category of coherent $\mathcal{O}_C$ -modules is equivalent to the category of finitely generated A -modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group of flat finitely generated A -modules $K_0(A)$. The structure theorem of modules over Dedekind domains states $K_0(A) \cong \mathbb{Z} \oplus \text{Cl}(A)$, where if M is any flat finitely generated A -module, then $M \cong A^r \oplus I$, where K is the fraction field of A , $r + 1 = \dim_K M \otimes_A K$ is the rank of M , and I is some fractional ideal in $\text{Cl}(A)$. If $N \cong A^s \oplus J$ is another A -module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\text{Cl}(A)$. $\square$

17. (a) Let $\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^3$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ . Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C , so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m .
- (b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.